Anastasija Ljubicic

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EDUCATION

Bachelor of Software Engineering *Oct 2023- Jun 2027 (expected)*

Faculty of Electrical Engineering - University of Belgrade (GPA 8.36)

TECHNICAL SKILLS

Languages:	Python, C, C++, Java, C#, JavaScript, Go
Web / Backend:	Django, Flask, SQLAlchemy, JWT, HTML, CSS, Bootstrap 5, JavaScript/AJAX, MVC
Databases:	SQL, MySQL, MongoDB, MS SQL Server, JDBC, Relational DB Design, XML/XQuery
ML / AI:	Neural Networks, CNNs, Fuzzy Logic
CS Fundamentals:	Data Structures & Algorithms, OOP, Design Patterns, Software Architecture, Operating Systems **,** Concurrency, RPC, Distributed Computing

PROJECTS

Systems Programming - *Simplified CLI* *C++(OOP)*

- Built a simplified CLI interpreter with support for command parsing, pipes, and file I/O redirection, implementing commands such as echo, time, date, touch, wc, tr, head, rm, truncate, prompt, and batch.

Distributed Systems – *Raft Consensus Algorithm* Go, RPC, Concurrency

- Independently implemented Raft consensus (leader election, log replication, persistence) from the original Stanford paper by Ongaro & Ousterhout, inspired by MIT's 6.5840 Distributed Systems coursework; validated with 10 self-written tests covering leader failure, network partitions, and quorum loss.

Database Systems – *SQL Query Optimizer* Python, SQL, Query Optimization, Algorithms, JSON

- Built a cost-based SQL query optimizer that parses and validates queries, generates evaluation plans, and estimates execution cost in block transfers.
- Implemented nested-loop, block/index nested-loop, sort-merge and hash joins, B+ tree and hash indexes, and external merge sort under configurable buffer constraints.

Cloud & Distributed Systems - *Investment Fund* Python/Flask, MongoDB, Redis, Docker, Kubernetes, Ethereum

- Built a containerized microservices investment platform with REST APIs, JWT authentication, MongoDB, Redis, and Kubernetes. Implemented persistent storage, three service replicas, and Ethereum smart-contract voting with majority consensus

Full-stack Web Development - *Compass* *Python/Django, MySQL, JS/AJAX, Git*

- Full-stack web platform for assessing student readiness for IT careers, featuring skill-gap analysis, internship listings, and personalized career path recommendations. Developed in a team of 4 across all SDLC phases - requirements, prototyping, database modeling, implementation, UML modeling, and unit/Selenium testing.

Database Systems – *Movie Rating Platform* Java, MS SQL Server, JDBC/JPA, SQL

- Movie rating platform with personalized recommendations and genre-based user profiling; implemented rating validation, reward assignment, and automatic trend detection via SQL stored procedures and triggers.

EXTRACURRICULAR ACTIVITIES

Fintech AI Hackathon - GARAZA startap community (*participant*)

Hackathon 2026 | AML Sanctions Screening Platform

- Collaborated on a three-layer AI-powered AML platform for screening global SWIFT transfers, combining traditional sanctions screening, an OSINT AI agent for investigating flagged individuals, and a predictive layer that flags high-risk countries ahead of official sanctions lists.

MICROSOFT SINERGIJA 2026 CONFERENCE *(event operations team)*

Electrical Circuits for Industry 4.0 - EESTech Challenge (*2nd place*)

International Hackathon 2025 | Smart Home Thermostat solution (backend & frontend & hardware)

- Owned backend architecture and core system logic, developed in Python on a Raspberry Pi 3.

Development Team Member of the Game Jam Hackathon 2025 - EESTEC LC Belgrade (organizing team)

Hackathon 2025 | Starter asset development for competition participants

- Developed reusable Unity/C# starter assets (player controllers, physics, spawners, camera systems, day-night cycle) and built playable prototypes (3D car driving, shooter, hurdle jumper) for competition participants.

LANGUAGES:

Languages: Serbian (native), English (Fluent), German (elementary)